

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 315 — Past Questions & Materials
by Renance

Course: MTH 315
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 7
Levels: 300

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Paper 1 of 7 — MTH 315 Vector & Tensor Analysis

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN.
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS
FIRST SEMESTER EXAMINATION 2024/2025 SESSION
COURSE TITLE & CODE: VECTOR AND TENSOR ANALYSIS (MTH 315)

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Instruction: Attempt any four questions Time: 2 hrs 30mins

Question 1

A particle moves along the curve $x = 2t^2, y = t^2 - 7t, z = 6t - 5$, where t is the time.

- Find the component of its velocity at time $t = 2$ in the direction $3i - 3j + 2k$. (7 marks)
- Find the component of its acceleration at time $t = 2$ in the direction $3i - 3j + 2k$. (3 marks)
- Find the directional Derivative of $\phi = x^2yz + 4xz^2$ at $(2, -4, -2)$ in the direction $4i - 2j - 4k$. (7.5marks)

Question 2

If $A = (2y + 3)i + xzj + (yz - x)k$, evaluate $\int_C A \cdot dr$ along the following paths C:

- $x = 2t^2, y = t, z = t^3$ from $t = 0$ to $t = 1$ (4 marks)
- the straight lines from $(0,0,0)$ to $(0,0,1)$, then to $(0,1,1)$ and then to $(2,1,1)$, (8.5marks)
- the straight lines joining $(0,0,0)$ and $(2,1,1)$ (5 marks)

Question 3

- Show that $F = (2xy + z^3)i + x^2j + 3xz^2k$ is a conservative force field (4.5marks)
- Find the work done in moving an object in this field from $(1, -2, 1)$ to $(3, 1, 4)$. (4 marks)
- If $A = 5t^2i + tj - t^3k$ and $B = \sin t i - \cos t j$, find $\frac{d^2}{dt^2}(A \times B)$. (9 marks)

Question 4

- If $\phi(x, y, z) = xy^2z$ and $A = x^2zi - x^3y^2j + yz^2k$, find $\frac{\partial^3}{\partial x^2 \partial z}(\phi A)$ at the point $(2, -2, -1)$. (6 marks)
- If $A = \cos xy i + (3xy - 2x^2)j - (3x + 2y)k$, find (i) $\frac{\partial^2 A}{\partial x^2}$ (ii) $\frac{\partial^2 A}{\partial x \partial y}$ (iii) $\frac{\partial^2 A}{\partial y^2}$ (6 marks)
- Evaluate $\int_{(0,0)}^{(2,1)} (10x^4 - 2xy^3)dx - 3x^2y^2dy$ (5.5marks)

Question 5

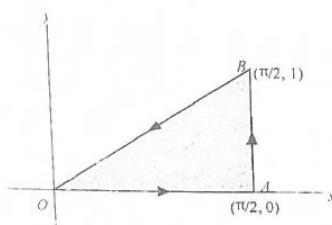
If $A = 2yz i - x^2y j + xz^2 k, B = x^2 i + yz j - xy k$ and $\phi = 2x^2yz^3$, find

- $A \cdot \nabla \phi$ at the point $(2, -2, -1)$ (4.5 marks)
- $B \cdot (\nabla \phi \times A)$ at the point $(-1, -2, 1)$ (5.5 marks)
- $(A \times B) \times \nabla \phi$ at the point $(1, 1, -1)$. (7.5 marks)

Question 6

Evaluate $\oint_C (y - \sin x)dx + \cos x dy$ where C is the triangle in the figure below

- directly (7 marks)
- by using Green's theorem in the plane. (7 marks)
- Show that the area bounded by a simple closed curve C is given by $\frac{1}{2} \oint_C x dy - y dx$ (3.5 marks)



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Paper 2 of 7 — MTH 315 VECTOR AND TENSOR ANALYSIS 2024/2025

Level 300 • First Semester • PastQuestion

in

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN.

COLLEGE OF SCIENCE :

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATION 2024/2025 SESSION - ONE

COURSE TITLE & CODE: VECTOR AND TENSOR ANALYSIS (MTH 315) ~

Instruction: Attempt any four questions Time: 2 hrs 30mins

Question 1

A particle moves along the curve $x = at^2 - 7t, z = 6t - 5$, where t is the time.

- Find the component of its velocity at time $t = 2$ in the direction $3i - 3j + 2k$. (7 marks)
- Find the component of its acceleration at time $t = 2$ in the direction $3i - 3j + 2k$. (3 marks)
- Find the directional Derivative of $\phi = xyz + 4xz^2$ at $(2, -4, -2)$ in the direction $4i - 2j - 4k$. (7.5 marks)

Question 2

$\mathbf{A} = (2y + 3)\mathbf{i} + xz\mathbf{j} + (yz - x)\mathbf{k}$, evaluate $\oint_C \mathbf{A} \cdot d\mathbf{r}$ along the following paths C:

- $x = 2t^2, y = t, z = t^3$ from $t = 0$ to $t = 1$ (4 marks)
- the straight lines from $(0, 0, 0)$ to $(0, 0, 1)$, then to $(0, 1, 1)$ and then to $(2, 1, 1)$, (8.5 marks)
- the straight lines joining $(0, 0, 0)$ and $(2, 1, 1)$ (5 marks)

Question 3,

- Show that $\mathbf{F} = (2xy + z^2)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$ is a conservative force field (4.5 marks)
- Find the work done in moving an object in this field from $(1, -2, 1)$ to $(2, 1, 4)$. (4 marks)
- If $\mathbf{A} = 5t^2\mathbf{i} + t\mathbf{j} - t^2\mathbf{k}$ and $\mathbf{B} = \sin t\mathbf{i} - \cos t\mathbf{j}$, find $\mathbf{A} \times \mathbf{B}$. (9 marks)

Question 4,

- If $(x, y, z) = xyz$ and $\mathbf{A} = x^2z\mathbf{i} - x^3y\mathbf{j} + y^2z\mathbf{k}$, find $\nabla \cdot \mathbf{A}$ at the point $(2, -2, -1)$. (6 marks)

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- If $\mathbf{A} = \cos xy\mathbf{i} + (3xy - 2x^2)\mathbf{j} - (3x + 2y)\mathbf{k}$, find $\nabla \cdot \mathbf{A}$ at the point $(2, 1, 1)$. (6 marks)

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Question 5

If $\mathbf{A} = 2yz\mathbf{i} - x^2y\mathbf{j} + xz^2\mathbf{k}$, $\mathbf{B} = x^2\mathbf{i} + yz\mathbf{j} - xy\mathbf{k}$ and $\phi = 2x^2yz$, find

- $\mathbf{A} \cdot \nabla \phi$ at the point $(2, -2, -1)$ (4.6 marks)
- $\nabla \cdot (\mathbf{A} \times \mathbf{B})$ at the point $(-1, -2, 1)$ (5.5 marks)
- $(\mathbf{A} \times \mathbf{B}) \cdot \nabla \phi$ at the point $(1, 1, -1)$. (7.5 marks)

Question 6

Evaluate $\oint_C (y - \sin x)dx + \cos x dy$ where C is the triangle in the figure below

(a) directly (7 marks)

(b) by using Green's theorem in the plane. (7 marks)

(c) Show that the area bounded by a simple closed curve C is given by $\oint_C x dy = \oint_C y dx$ (3.5 marks)

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Paper 3 of 7 — MTH 315 Vector and Tensor 2019 2020 First Semester

Level 300 • Second Semester • PastQuestion

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MARKING SCHEME

70%



FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

BSc (MATHEMATICS) DEGREE EXAMINATION (2019/2020), SECOND SEMESTER.

MTH 315 (Vector and Tensor) 25th November, 2021.

INSTRUCTIONS;

- Answer FOUR questions Only.
- Write the number of the question on the answer sheet
- All answer must be written on the answer sheets provided
- Duration: $2\frac{1}{4}$ hrs
- Total examination score 70%

Question 1. Define the following terms in vector (i) Modulus of a vector and Unit vector given

$$a = a_1i + a_2j + a_3k$$

(2mks)

b. Given that $a = 1i + 2j - 2k$. Find the:

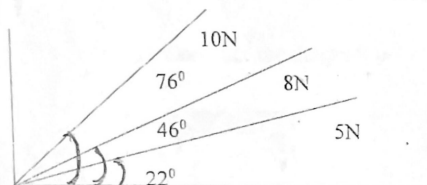
- (i) The Mag of a
- (ii) the unit vector in the direction of a
- (iii) The direction cosine of a
- iv) the angle between 3a-v

(15 $\frac{1}{2}$ mks)

Question 2a. Show that the set of vectors;

r_1, r_2 and r_3 given by $r_1 = j - 2k, r_2 = i - j + k$ and $r_3 = i + 2j + k$ is linearly dependent? (5mks)

b. Calculate the resultant of these three force 10N, 8N and 5N acting on a body as shown in the diagram.



(12 $\frac{1}{2}$ mks)

Question 3. If r_1, r_2 and r_3 are vectors. State all the Rules of differentiation of the vector (3mks)

b. If $r = x^2 \sin y i + x^2 y^2 \cos y j - C \log \left(\frac{1}{x} + \sqrt{x^2 y - c} \right) k$. Find $\frac{\partial r}{\partial x}, \frac{\partial r}{\partial y}, \frac{\partial^2 r}{\partial x \partial y}$ and $\frac{\partial^2 r}{\partial y^2}$ (14 $\frac{1}{2}$ mks)

Question 4. If r and s are vectors. State three integral value of vectors r and s (3mks)

b. Solve for r when the equation $a \times 2 \frac{d^2 y}{dt^2} = b$ where a and b are constant. (5mks)

c. Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F}(x, y, z) = e^{2x}i + z(y+1)j + z^3k$ and C is given

$$\text{by } \vec{r}(t) = t^3i + (1-3t)j + e^t k \text{ for } 0 \leq t \leq 2$$

(9 $\frac{1}{2}$ mks)

Question 5 Given that $A(w) = 3w^3 - 2w^2 - w)i + (3w+3)j + (10w^3 - 3w + 12)k$ (8mks)

$$\text{Evaluate } \int_1^2 A(w)dw$$

$$\text{b. Evaluate } \int_1^3 A \cdot B \times C(w)dt.$$

(9 $\frac{1}{2}$ mks)

$$\text{Where } A = t^2i + 3(3t^2 + 1)j + 2k, B = 3t^2i + 3j + 4k \text{ and } C = 3it + j + tk$$

Question 6. Define Tensor and state the mathematical equation between Dyadic and Triadic equations (4mks)

b. Let $r(x, y, z) = xi + xj + zk$ be the position vector field on $\mathbb{R}^4$.

$$\text{The } \|r(x, y, z)\|^4 = r \cdot r \cdot r \cdot r = x^4 + y^4 + z^4 \text{ is a real -value function.}$$

Find

(i) Gradient of $\|r\|^4$ (ii) The divergent of r

(iii) The curl of r iv) the Laplacian of $\|r\|^4$

(14 $\frac{1}{2}$ mks)

Paper 4 of 7 — MTH 315 Vector and Tensor Analysis 2018 2019 First Semester

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN,.

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

FIRST SEMESTER EXAMINATION 2018/2019 SESSION

COURSE TITLE: VECTOR AND TENSOR ANALYSIS

COURSE CODE: MTH 315

four questions Time: 3 hours

Instruction: Answer any

Question One

- a) Compute the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ from $(0,0,0)$ to $(1,2,4)$ if $\mathbf{F}$ along the line segment joining these two points given that $\mathbf{x} =$
- (b) Show that $\mathbf{F}$ is a conservative

$$\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + (xz - y)\mathbf{k}$$

$$t, y = 2t, z = 4t^3$$

€ force field. If $\mathbf{F} = (2xy + 23)\mathbf{i} + (x^2)\mathbf{j} - 3xz^2\mathbf{k}$

- (c) Find the scalar function ϕ such that $\mathbf{F} = \nabla\phi$. If $\mathbf{F} = (2xy + 23)\mathbf{i} + (x^2)\mathbf{j} - 3xz^2\mathbf{k}$

Question Two

- (a) State the Gauss' divergence theorem
- (b) Verify divergence due to Gauss for $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + 2z\mathbf{k}$ taken over a cube $0 < x, y, z < 1$
- (c) Calculate the volume of a tetrahedron, whose vertices are at the points $A(1,2,-1)$, $B(2,0,1)$, $C(1,1,2)$, $D(2,4)$.

Question Three

- (a) If $\mathbf{A} = x^2\mathbf{i} - 2x\mathbf{j} + 2yz\mathbf{k}$ and $\phi = 3x^2y + z^2$ Find $\nabla \cdot \mathbf{A}$, $\nabla \times \mathbf{A}$ and $\text{Div}(\text{grad } \phi)$

$$-2$$

$$\nabla \cdot \mathbf{r}$$

- (b) Show that $\nabla \cdot \mathbf{r} = n$

- (c) Given that $\mathbf{V} = (3x + y)\mathbf{i} + (x - y)\mathbf{j} + (x^2 - 2az)\mathbf{k}$ what is $\mathbf{V}$ if $\mathbf{V}$ is solenoidal?

Question Four

- (a) Let $\mathbf{F} = (x^2 + y^2)\mathbf{i} + (3x^2y)\mathbf{j}$, Find the line integral of the tangential component of $\mathbf{F}$ from $(-1,0)$ to $(1,0)$ along the semi-circle $y = \sqrt{1 - x^2}$

$$1,0) \text{ to } (1,0) \text{ along the semi-circle } y = \sqrt{1 - x^2}$$

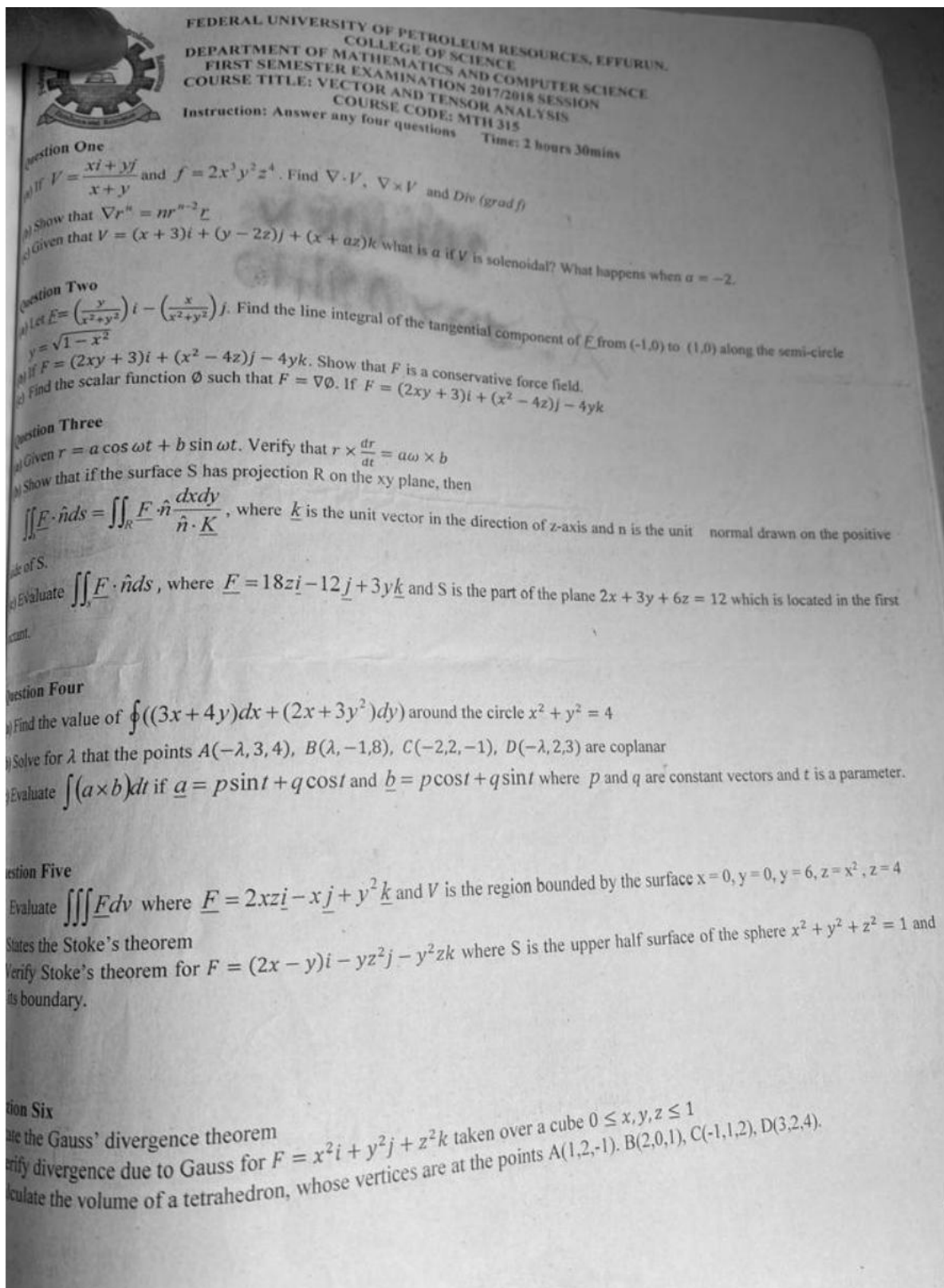
- (b) Show that the points $A(-1, 3, 4)$, $B(1, 8)$, $C(-2, 2, -1)$, $D(-1, 2, 3)$ are coplanar

(c) Evaluate $\int (ax+b) \sqrt{a^2x^2+b^2} dx$ if $a = p \sin t + q \cos t$ and $b = p \cos t + q \sin t$ where p and q are constant
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 vectors and t is a parameter.
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Paper 5 of 7 — MTH 315, 2018.

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Paper 6 of 7 — MTH 315, 2020 past questions

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said to be linearly dependent

is the vector linearly independent or not

$= 3a - 5b + 2c$. % $= 4a - 5b + C$

\$.5mks

2mks

+ Scand rz = @ show, mks

iven by, $= 2a - 3b + 6, 72$

janar vector is 'linearly dependent.

acting at a point mks

es for differentiation of these vectors 3mks

12.5mks

ectors rand S- 2mks

Smks

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vector field on BR. Then $\nabla \cdot \mathbf{r}$ at y,

Paper 7 of 7 — MTH 315, 2019

Level 300 • First Semester • PastQuestion

$2x^2 y + 2yz^2 =$

$\nabla \cdot \mathbf{A} = 3x^2 y^3 z^2$ Find $\nabla \cdot \mathbf{A}$, $\nabla \times \mathbf{A}$ and $\text{Div}(\text{grad } \phi)$

$\mathbf{v} = \nabla r$

$\mathbf{V} = A \sin 2az$ what is $\mathbf{A}$ if $\mathbf{V}$ is solenoidal?

Find the line integral of the tangential component of $\mathbf{F}$ from $(-1, 0, 0)$ to $(1, 0, 0)$ along the semi-circle $\mathbf{r} = \cos \theta \mathbf{i} + \sin \theta \mathbf{j}$

$\mathbf{r} = \cos \theta \mathbf{i} + \sin \theta \mathbf{j}$

along the semi-circle $\mathbf{r} = \cos \theta \mathbf{i} + \sin \theta \mathbf{j}$

the points $A(-1, 3, 4)$, $B(1, 1, 8)$, $C(2, 2, 2)$, $D(-1, 2, 3)$

$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}$ where $\mathbf{a}$ and $\mathbf{b}$ are vectors

are coplanar

constant